DESIGN AND NUMERICAL ANALYSIS OF A SUB-SURFACE TWO-BODY WAVE ENERGY CONVERTOR

Virag Mishra, Michael Kelly, Joseph Piacenza, and Solomon C. Yim

***Abstract*—Recent developments in self-reacting oscillating Wave Energy Convertors (WECs) point toward two popular design concepts. These designs are either semi-submerged system in which a surface piercing float reacts against a relatively still heave plate or a single body system that reacts against the ocean floor. The first design employs a rigid spar connection between the actuator and the reaction body; however, the second design uses a flexible belt connection between the actuator and the foundation. In this work, a two-body sub-surface WEC is proposed in which the submerged float reacts against a base that is dynamically active. The dynamics of the reacting body is influenced by both the buoyancy parameters and the PTO generated tether tension. As a result, the two-body system oscillates in an antipodal mode. The optimal hydrostatic conditions that result in a design that can generate mechanical power, in single degree of freedom motion, under the wave excitation are discussed in this work. The hydrodynamic analysis is carried out for different wave conditions observed off the coast of US. Optimal PTO parameters are determined for various designs. The analysis is carried out using Ansys-Aqwa. Observations of the relative predictive performance between the two numerical simulations models are delineated.**

***Index Terms*—Hydrodynamics, Power Take Off (PTO), Wave Energy Converter (WEC)**

I. INTRODUCTION

DUE to the concerted effort to achieve energy independence as well as to reduce the greenhouse gas emission, most of the development to date in offshore renewable energy systems has been targeted at "utility-scale" or "community-scale" devices. These devices have electricity generation capacities in the order of MW and kW respectively. However, another application area that is attracting investment in wave energy converters is "small-scale" devices. These devices can provide autonomous power to various ocean-bound systems, such as autonomous underwater vehicles [1], observation systems [2], and even aquaculture farms. The expectation is that these devices can be used to power the on-site stand-alone systems which are located remotely, resulting in exponential growth in the sector often referred as "Blue economy" [3], [4]. This paper provides an overview of the initial design process of such a device. First, the primary dimension of the device is derived based on the constraints of the operating design (Sect. yyy). The details of the deployment sites are discussed in (Sect. xxxxxx). The results and the following discussion are presented in (Sect. zzzz). This project is funded by the NAVSEA, as a one-year grant to design a sub-surface WEC capable of continuous power generation of 100W at various sites with distinct wave characteristics off the coast of the United States.

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II. BACKGROUND

Numerical benchmarking of eight WECs with different working principles was conducted to evaluate their power matrices and site specific annual energy production [5]. Out of these, two operated close to seabed and only one was near surface submerged WECs. This shows that the focus has mainly been around floating devices. Some of the popular floating WECs with heave plate are MBARI-WEC [6], OPT Powerbuoy, OSU-Lupa (Laboratory Upgrade Point Absorber) [7] and Wavebob [8].

Though the wave energy decays exponentially with depth in deep water regime, subsurface WECs provide the potential benefit of survivability and covertness. Possibly these are reasons why subsurface WECs are emerging as a preferred design choice for WEC developers. Some of the popular designs that are either bottom fixed or do not have a heave plate as reacting bodies are AWS Ocean Energy's Archimedes Waveswing, Symphony Wave Power's WEC, Carnegie Clean Energy's CETO 6 [9], and CalWave's xWave. CETO6 and xWave are multi tethered, multi DOF point absorbers which operate close to the surface. The Waveswing is an air pressure differential point absorber tethered to the sea bed and Symphony Wave Power's WEC is a bottom-fixed device operating in shallow waters. Conceptual submerged WEC capable of producing 10 W of power with a float and six trubines has been proposed in [10].

In all the buoyant designs mentioned above the buoyancy of the float acts as the actuation force and either the foundation in the sea bed or the heave plate acts as a reaction body. In this work we propose a two-body submerged system with a float and heave plate connected by a tether and the entire system is moored to the foundation. The float will act as an actuator and

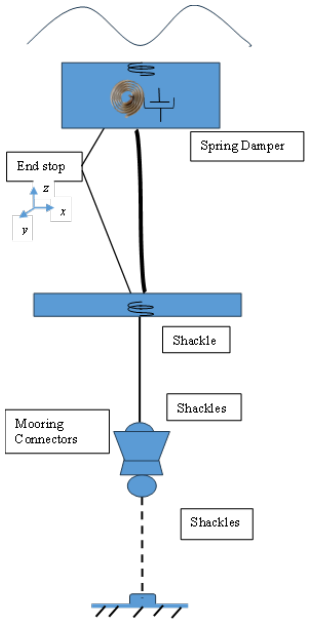


Figure 1. Sketch of the two-body sub-surface WEC under analysis

the buoyancy of the heave plate will aid to power generation. This work will evaluate the performance for the above described WEC and compare six different design candidates at different wave energy sites in the US.

III. Device Details

The WEC design under study is a two-body point absorber consisting of a submerged float and a heave plate. The design of the WEC supports the sub-surface operation. The sketch of primary components used in the numerical modelling are shown in Fig. 1. The float and heave plate are connected by a tether which is attached to the top of the heave plate and the bottom of the float. The heave plate is moored to the sea bed using a mooring line. The mooring line is attached to the bottom of the heave plate and the other end is attached to a foundation on the sea bed. The tether and mooring line are assumed to be linear elastic with a stiffness of k and $0.1k$ respectively. The tether and mooring line are assumed to be massless and inextensible.

To determine optimal device configuration multiple designs were analyzed. In this paper preliminary results for some of the designs considered are discussed and compared. The relative dimensions of the float and Heave Plate for various designs considered are shown in Fig. 2. A summary of the important dimensions and characteristics of these designs are presented in Table I. R_f and R_{hp} are the radius of float and heave plate respectively and H_f, H_{hp} are the height of float and heave plate. Similarly, W_f and W_{hp} are the weight of the float and heave plate. The primary difference between various designs is that the total weight of the design D0 was 8000 kg, however the new designs are comparatively lighter in the range 2100 kg (D1, D2), 2500 kg (D3, D4) and (D5, D6) which is 2966 kg. The tether tension in the design D0 was 23696 N, which would increase the CAPEX cost as the cost is proportional to the tether cost. However, the tether tension in other designs is significantly lower.

IV. Numerical Modelling

Commercially available software "Ansys-Aqwa®" was used to both calculate the frequency dependent parameters and to simulate the time-dependent dynamic behavior of the WEC.

A. Cable modelling

A linear elastic cable module in Aqwa is the simplest numerical model of an actual mooring line. Aqwa requires the user to specify the unstretched length instead of pretension for elastic tethers. With the knowledge of stiffness and pretension (calculated as the difference between the buoyancy and the weight of the float and Heave Plate), the stretched length can be determined.

$$L^t = \frac{(V_f - V_f^{disp})\rho_w g}{k} \tag{1}$$

$$L_0^t = D_{hp} - (H_f + D_f) - L^t \tag{2}$$

$$L^m = \frac{(V_f + V_{hp}) - (V_f^{disp} + V_{hp}^{disp})\rho_w g}{0.1k} \tag{3}$$

$$L_0^m = D - (D_{hp} + H_{hp}) - L^m \tag{4}$$

The stretched length L^t of the tether and mooring can be calculated using (1) and (3). The initial unstretched length L_0 of the tether and mooring can be calculated using (2) and (4) respectively. V_x^{disp} is the displaced volume of the respective cylindrical geometry (x can be either "f" for float or "hp" for heave plate.), ρ_w is the density of water, g is the acceleration due to gravity, k is the stiffness of the tether, D_x is the distance from the free surface to the respective cylindrical geometry, H_x is the height of the respective cylindrical geometry. D is the depth of water at deployment location (considered $= 70$ m for all the simulations).

B. PTO algorithm

The PTO system is modelled as a linear spring-damper system with a stiffness K_{mech} and p C_{mech}. The mechanical spring force is given by:

$$F_{mech} = C + K_{mech}\Delta L \tag{5}$$

V. Wave Climate at Deployment Locations

Based on the works of [11] on wave resource assessment off the coast of US North West, a set of guidelines were developed and included in a technical specification for wave resource assessment by International Electrotechnical Commission (IEC) [12]. The IEC brings together international marine energy experts to develop consensus-based standards that guide the development of a broad range of sectors. The current industry consensus is determination of representative sea states based on the energy weighted joint probability distribution function of primarily H_s and T_e/T_P [13]. However, the true description of a sea state should involve additional parameters such as spectral shape, peak direction, directional spreading, wind, tide, and

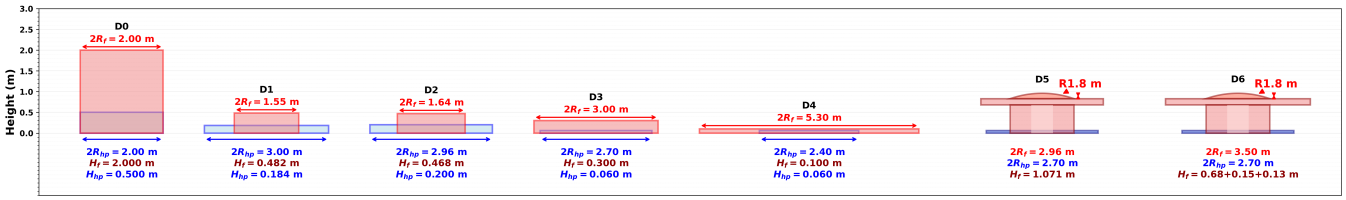


Figure 2. Representative dimensions and designs of the float and heave plate for hydrodynamic analysis. The heave plate and float were connected to the tether not shown in the above image.

Table I
DESIGN PARAMETERS AND LOADING CHARACTERISTICS OF THE CANDIDATE WEC CONFIGURATIONS.

Design	W_f **(kg)**	W_{hp} **(kg)**	$W_f + W_{hp}$ **(kg)**	R_f **(m)**	R_{hp} **(m)**	H_f **(m)**	H_{hp} **(m)**	**Tether tension (N)**	**Mooring tension (N)**
D0	4000	4000	8000	1	1	2	0.5	23931	201
D1	700	1400	2100	0.777	1.5	0.482	0.184	2325	1601
D2	800	1600	2400	0.82	1.48	0.468	0.2	2092	172
D3	1500	1000	2500	1.5	1.35	0.3	0.06	6606	203
D4	1500	1000	2500	2.65	1.2	0.1	0.06	7466	358
D5	2066	900	2966	1.48 (max)	1.35	1.027 (total)	0.06	5782	289
D6	2066	900	2966	1.75 (max)	1.35	0.853 (total)	0.06	5782	277

current; yet data to fully characterize these aspects are often unavailable. Even when such data exist, current approaches neither develop a multidimensional probabilistic description of the deployment site nor account for energy-weighted probabilities, and they typically preclude the evaluation of WEC performance across multiple deployment climates. A need for developing a methodology that could account for multiple deployment locations was presented in a Wave Energy Prize competition by US Department of Energy in 2016. A new methodology was proposed in this competition by [14] that applied a cluster analysis to scatter diagrams in order to determine a subset of sea states that could be scaled to find the average annual power flux at each wave climate. A common centroid representing of H_s and T_P was determined for the clusters. This procedure opens the possibility of using a single set of cluster centroid for different locations. Since WEC performance assessment simulations are computationally resource intensive, in this work the methodology described above is used to run simulations for a single set of cluster centroids and the power produced for each centroid is scaled depending on the site under consideration.



Figure 3. The sea states used for the analysis of WEC performance

A set of 18 sea states, shown in Fig. 3, was used throughout the analysis in this report. Wave elevations were generated using the Bretschneider spectrum. Although these wave datasets do not all correspond to the same water depth (ranges from 42 m to 149 m), they provide a useful indication of WEC performance across varied wave profiles, since the same set of sea states is applied at each location shown in Fig. 4. Note that when the weights associated with the 18 sea states are used, they account for varying percentages of the total joint probability distribution (67%–95%), depending on location. This is because, at some locations, a significant portion of the year falls into sea states not represented within the 18 sea states considered.



Figure 4. Location of the NDBC buoys used for the analysis of wave climate

Site specific optimal PTO coefficient for the WEC are determined for each location shown in Fig. 4. These sites either host a testing facility or a deployment berth and are probable sites for the development of WEC farms. The significant wave height reported by NDBC (and used here) is the average of the highest one-third of all the wave heights during a 20-minute sampling period. The peak period is the period with maximum energy in the spectrum. The list of the buoy locations chosen for the analysis is presented in Table II.

Occurrence data reported by NDBC buoys were used to calculate the annual average power at that location. A cluster analysis was applied to wave scatter diagrams to identify a representative subset of sea states

Table II
NDBC BUOYS CHOSEN FOR ANALYSIS

Buoy ID*	US Location	Depth [m]	Year
51207	Oahu, HI	81	2019
42001	Mid Gulf, New Orleans, LA	32	2018
46050	PACWAVE, OR	149	2017
46254	San Diego, CA	46	2019
41009	Canaveral, FL	42	2011
44018	Cape Cod, MA	44	2010

* All wave data comes from National Data Buoy Center (NDBC)

that can be scaled to estimate the average annual power (AAP) flux. The subset is determined by minimizing the distance between each data point and its corresponding cluster centroid. Four pdist2 options for minimizing the distance function were compared these are Squared Euclidean, Standardized Euclidean, Eclidean and Mahalanobis. This clustering is performed using the k-means algorithm implemented in the MATLAB Statistics Toolbox, where the number of clusters and the centroid values is specified by the user. The algorithm employs an iterative partitioning procedure that optimizes the associated memberships by minimizing the sum of point-to-centroid distances across all k clusters. As a result, each sea state is assigned to the cluster whose distance to centroid yields the minimum value, providing an optimally representative classification of the wave climate. In this paper results corresponding to "Mahalanobis" method are presented.

VI. Some common mistakes

The word 'data' is plural, not singular. A graph within a graph is an 'inset,' not an 'insert.' The word 'alternatively' is preferred to the word 'alternately' (unless you really mean something that alternates). Use the word 'whereas' instead of 'while' (unless you are referring to simultaneous events). Do not use the word 'essentially' to mean 'approximately' or 'effectively.' Do not use the word 'issue' as a euphemism for 'problem.' Be aware of the different meanings of the homophones 'affect' (usually a verb) and 'effect' (usually a noun), 'complement' and 'compliment,' 'discreet' and 'discrete,' 'principal' (e.g., 'principal investigator') and 'principle' (e.g., 'principle of measurement'). Do not confuse 'imply' and 'infer.' Prefixes such as 'non,' 'sub,' 'micro,' 'multi,' and 'ultra' are not independent words; they should be joined to the words they modify, usually without a hyphen. There is no period after the 'et' in the Latin abbreviation 'et al.' (it is also not italicized). The abbreviation 'i.e.,' means 'that is,' and the abbreviation 'e.g.,' means 'for example' (these abbreviations are not italicized). The suffix '-ize' is correct, both phonetically and etymologically, in both British and American English and there is no reason to use '-ise.' For further information, please see the relevant entry in the Oxford English Dictionary.

VII. Guidelines for graphics preparation and submission

A. Types of graphics

The following list outlines the different types of graphics published in IMEJ journals. They are categorized based on their construction, and use of colour or shades of grey:

1) Colour/grey-scale figures: Figures that are meant to appear in colour, or shades of black/grey. Such figures may include photographs, illustrations, multicolour graphs, and flowcharts.

2) Line art figures: Figures that are composed of only black lines and shapes. These figures should have no shades or half-tones of grey, only black and white.

3) Tables: Data charts which are typically black and white, but sometimes include colour.

4) Multipart figures: Figures compiled of more than one sub-figure presented side-by-side, or stacked. If a multipart figure is made up of multiple figure types (one part is line-art, and another is grey-scale or colour) the figure should meet the stricter guidelines.

B. File formats for graphics

Format and save your graphics using a suitable graphics processing program that will allow you to export images as PostScript (PS), Encapsulated PostScript (.EPS), Tagged Image File Format (.TIFF), Portable Network Graphics (.PNG) or JPEG (.JPG) at suitable resolution (see below). It is strongly recommended that graphics be saved in PNG format rather than DOC, XLS, or PPT native formats. Doing so will protect your figures from common problems that occur when working on the files across multiple platforms. When submitting your final paper, your graphics should all be submitted individually in one of these formats along with the manuscript.

C. Sizing of graphics

Most charts, graphs, and tables are one column wide (86 mm) or page wide (178 mm). The maximum depth a graphic can be is 250 mm. When choosing the depth of a graphic, please allow space for a caption. Figures can be sized between column and page widths if the author chooses, however it is recommended that figures are not sized less than column width unless when necessary. IMEJ uses page size A4 with a column measurement of 86 mm. The column spacing is 6 mm. The left and right margins are 16 mm, top margin 22 mm (including header) and bottom margin 14 mm. The header height is 10 mm.

D. Raster graphics

The proper resolution of your figures will depend on the type of figure it is as defined in the 'Types of Figures' section. Photographs, and colour and grey-scale figures should be at least 300 dpi. Line art, including tables should be a minimum of 600 dpi. The JPEG format should only be used for photographs; PNG format should be used for diagrams as it preserves sharp boundaries.

E. Vector graphics

In order to preserve the integrity of figures across multiple computer platforms, we accept vector graphics in the following formats: .EPS/.PDF/.PS. All fonts must be embedded or text converted to outlines in order to achieve the best-quality results.

F. Colour space

The term colour space refers to the entire sum of colours that can be represented within the said medium. For our purposes, the three main colour spaces are: grey-scale, RGB (red/green/blue) and CMYK (cyan/magenta/yellow/black). RGB is generally used with on-screen graphics, whereas CMYK is used for printing purposes. Colour figures should be generated all in RGB or all in CMYK colour space. grey-scale images should be submitted in grey-scale colour space. Line art may be provided in grey-scale OR bitmap colour space. Note that 'bitmap colour space' and 'bitmap file format' are not the same thing. When bitmap colour space is selected, .PNG or .TIFF are the recommended file formats.

G. Accepted fonts within figures

When preparing your graphics IMEJ suggests that you use of one of the following fonts: Palatino, Times New Roman, Helvetica, Arial, Cambria, and Symbol. If you are supplying EPS, PS, or PDF files all fonts must be embedded. Some fonts may only be native to your operating system; without the fonts embedded, parts of the graphic may be distorted or missing. A safe option when finalizing your figures is to strip out the fonts before you save the files, creating 'outline' type. This converts fonts to artwork what will appear uniformly on any screen.

H. Using labels within figures

1) Figure axis labels: Figure axis labels are often a source of confusion. Use words rather than symbols. As an example, write the quantity 'Power,' or 'Power P,' not just 'P.' Put units in parentheses. Do not label axes only with units. As in 1, for example, write 'Flow speed (m/s)' or 'Flow speed (m·s^{-1}),' not just 'm/s.' Do not label axes with a ratio of quantities and units. For example, write 'Pressure (Pa),' not 'Pressure/Pa.' Multipliers can be especially confusing. Write 'Pressure (kPa)' or 'Pressure (10^3 Pa).' Do not write 'Pressure Pa×1000' because the reader would not know whether the top axis label in 1 meant 16000 Pa or 0.016 Pa. Figure labels should be legible, approximately 8 to 10 point type.

2) Subfigure labels in multipart figures and tables: Multipart figures should be combined and labelled before final submission. Labels should appear centred below each subfigure in 8 point Palatino font in the format of (a) (b) (c).

I. File naming

Figures (line artwork or photographs) should be named with the first author's surname, followed by the number that represents the sequential location of this image in your article. For example, in author "Smirnov's" paper, the first three figures would be named smirnov1.tif, smirnov2.tif, and smirnov3.ps. Tables should contain only the body of the table (not the caption) and should be named similarly to figures, except that '.t' is inserted in-between the author's name and the table number. For example, author "Nguyen's" first three tables would be named nguyen.t1.tif, nguyen.t2.ps, nguyen.t3.eps.

J. Referencing a figure or table within your paper

When referencing your figures and tables within your paper, use the abbreviation 'Fig.' even at the beginning of a sentence. Do not abbreviate 'Table.' Tables should be numbered with Roman numerals.

K. Submitting your graphics

Because IMEJ is run for and by the marine renewable energy community, without an external publisher and with only limited resources for graphical design, you are requested to position figures and tables at the top of each page, in the correct place in the document, as you would expect to see them in the final manuscript. In addition to submitting figures within your final manuscript, figures should be submitted individually, separate from the manuscript in one of the file formats listed above in section VI-C. Place figure captions below the figures; place table titles above the tables. Please do not include captions as part of the figures, or put them in 'text boxes' linked to the figures. Also, do not place borders around the outside of your figures.

L. Colour printing in IMEJ

As the IMEJ is an online-only journal, an author can publish colour figures on IMEJ at no charge. However, authors may wish to bear in mind that readers will often print journal articles in monochrome.

VIII. References and footnotes

A. References

In all references, the given name of the author or editor is abbreviated to the initial only and precedes the last name [15]. Use them all; use et al. only if names are not given [16]. Abbreviate conference titles. When referencing a patent, provide the day and the month of issue, or application. References should include enough information for the reader to find the original source. Do not combine references. There must be only one reference with each number [17]. If there is a URL included with the print reference, it can be included at the end of the reference [5], [17].

Capitalize only the first word in a title, except for proper nouns and element symbols. For papers published in translation journals, please give the English citation first, followed by the original foreign-language citation.

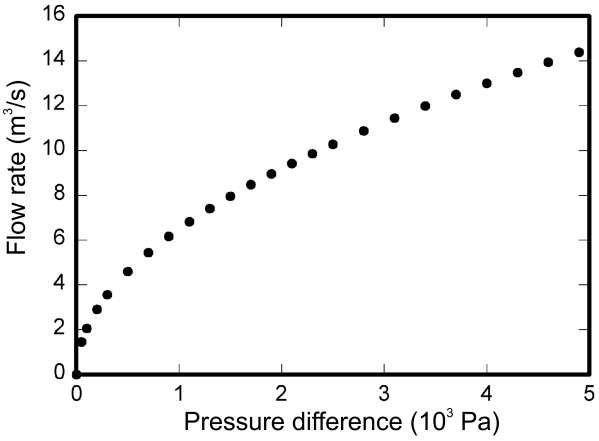


Figure 5. Flow rate as a function of pressure difference. Note that "Fig." is abbreviated. There is a period after the figure number, followed by two spaces. It is good practice to explain the significance of the figure in the caption.

Table III
EXAMPLE VARIABLES AND UNITS FOR MARINE RENEWABLE ENERGY

Symbol	Quantity	Unit
U	Flow speed	1 m/s = $0.51\dot{4}$ kn[a]
p	Pressure	Pa
μ	Dynamic viscosity	Pa·s
ν	Kinematic viscosity	m^2/s
L	Distance	1000 m = 1 km = $\frac{250}{463}$ nmi[a]
E	Energy	1 kJ = $0.27\dot{7}$ kWh
a	Acceleration	$m{\cdot}s^{-2}$
F	Force	N
P	Power	W
M	Mass	1000 kg = 1 t
Q	Flow rate	m^3/s

Vertical lines in tables are deprecated. Statements that serve as captions for the entire table do not need footnote letters.
[a] The Admiralty measured mile and Admiralty knot should not be used to avoid confusion.

B. Footnotes

It is recommended that footnotes be avoided (except for the unnumbered footnote with the receipt date on the first page). Instead, try to integrate the footnote information into the text.

IX. SUBMITTING YOUR PAPER FOR REVIEW

A. Review stage

Contributions to the IMEJ may be submitted electronically on IMEJ's online manuscript submission and peer-review system at marineenergyjournal.org. First check if you have an existing account. If there is none, please register a new account. After logging in, select "Submissions" and then "New Submission." Along with other information, you will be asked to confirm that the paper conforms to the publishing policy (see sections X and XI). There are various steps to the submission process; you must complete all steps for a complete submission. You may compress large files prior to upload using *7-Zip* or *Gzip*. At the end of each step you must click "Save and Continue"; just uploading the paper is not sufficient. After the last step, you should see a confirmation that the submission is complete. You should also receive an e-mail confirmation. For inquiries regarding the submission of your paper, please contact managing-editor@marineenergyjournal.org.

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XI. PUBLICATION PRINCIPLES

The IMEJ publishes peer-reviewed articles covering fundamental and applied research, including: case studies; field deployment and testing; technical, economic and social analysis; and laboratory based studies covering wave, tidal and ocean thermal energy conversion. Authors should consider the following points:

1) Technical papers submitted for publication must advance the state of knowledge and must cite relevant prior work.
2) The length of a submitted paper should be commensurate with the importance, or appropriate to the complexity, of the work. For example, an obvious extension of previously published work might not be appropriate for publication or might be adequately treated in just a few pages.
3) Authors must convince both peer reviewers and the editors of the scientific and technical merit of a paper; the standards of proof are higher when extraordinary or unexpected results are reported.
4) Because replication is required for scientific progress, papers submitted for publication must provide sufficient information to allow readers to

perform similar experiments or calculations and use the reported results. Although not everything need be disclosed, a paper must contain new, useable, and fully described information. Authors should expect to be challenged by reviewers if the results are not supported by adequate data and critical details.

5) Papers that describe ongoing work or announce the latest technical achievement, may not be appropriate for publication. However, they may suitable for presentation at the European Wave and Tidal Energy Conference (ewtec.org), for example.

XII. Discussion and Conclusion

This study investigated the hydrodynamic performance and design optimization of a submerged wave energy converter (WEC) equipped with a tether–PTO system. The report first introduced the overall system architecture, including the prime movers, tether specifications, PTO algorithm, and PTO parameters. A numerical modelling framework was then developed, covering the governing theory, model implementation, verification of the developed UDF and the selection of representative sea states for performance assessment. The mechanical average annual power capture at six different sites along the US coasts for different designs is compared.

A comprehensive sensitivity analysis was conducted under both survival and operational sea states. The operational-state analysis examined the influence of key design parameters for six different designs of float and heave plate. These designs vary in characteristic dimensions. Several sensitivity analyses were carried out for different operating parameters, primarily the PTO coefficients, draft, initial tether length, mass ratio between float and heave plate and mass of heave plate. The results demonstrate that the hydrodynamic response and power capture of the submerged WEC are strongly influenced by both the system configuration and the PTO parameters. The sensitivity analyses showed that careful selection of key design parameters, including the geometric configuration, mass distribution, and the initial float–heave plate gap, can substantially enhance energy capture while maintaining acceptable structural loads and system stability. For a configuration that is submerged 2 m below the surface, the power capture at most of the sites for most of the PTO coefficients is more than the minimum targeted value (only location where power was lower was in San Diego). The variation of power capture at various depths and different tether lengths is also compared for two designs. A linear drop in the captured power is observed with an increase in the draft and increase in the initial tether length (over the range of 2-7 m submerged). The optimal PTO coefficient at six different sites around US were found out for maximum power capture.

Overall, the validated numerical model and the comprehensive parametric analyses presented in this study provide a robust framework for optimizing submerged WEC designs, enabling an effective balance between energy production, structural integrity, operational reliability and survivability for site-specific deployment.

Appendix A
The Lanchester-Betz limit in a relativistic rotating reference frame

Appendices, if needed, appear before the acknowledgement.

Appendix B

The second appendix text goes here. You can choose not to have a title for an appendix by leaving the argument blank.

ACKNOWLEDGEMENT

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